Quantitative Methods I

Week 2 - More Functions

Jan Christopher Kops

QM I, SBE, Maastricht University

Motivation

There is more to business than last week's linear and quadratic functions.

Question	**Tool**
• How model fixed & per-unit costs?	→ Broken (Rational) Functions
• How relate input to output?	→ Power Functions
• How model growth?	→ Exponential & Log Functions
• How achieve desired output?	→ Inverse Functions

Broken (Rational) Functions

- In business, we often deal with ratios, like cost per unit, or revenue per employee. These can be modeled with **broken (rational) functions**.

- Any such function is the quotient (or ratio) of two polynomials:

$$f(x) = \frac{p(x)}{q(x)} \quad \text{(where } q(x) \neq 0\text{)}$$

- A key application is finding the minimum average cost, which is a classic optimization problem. Let's look at an example.

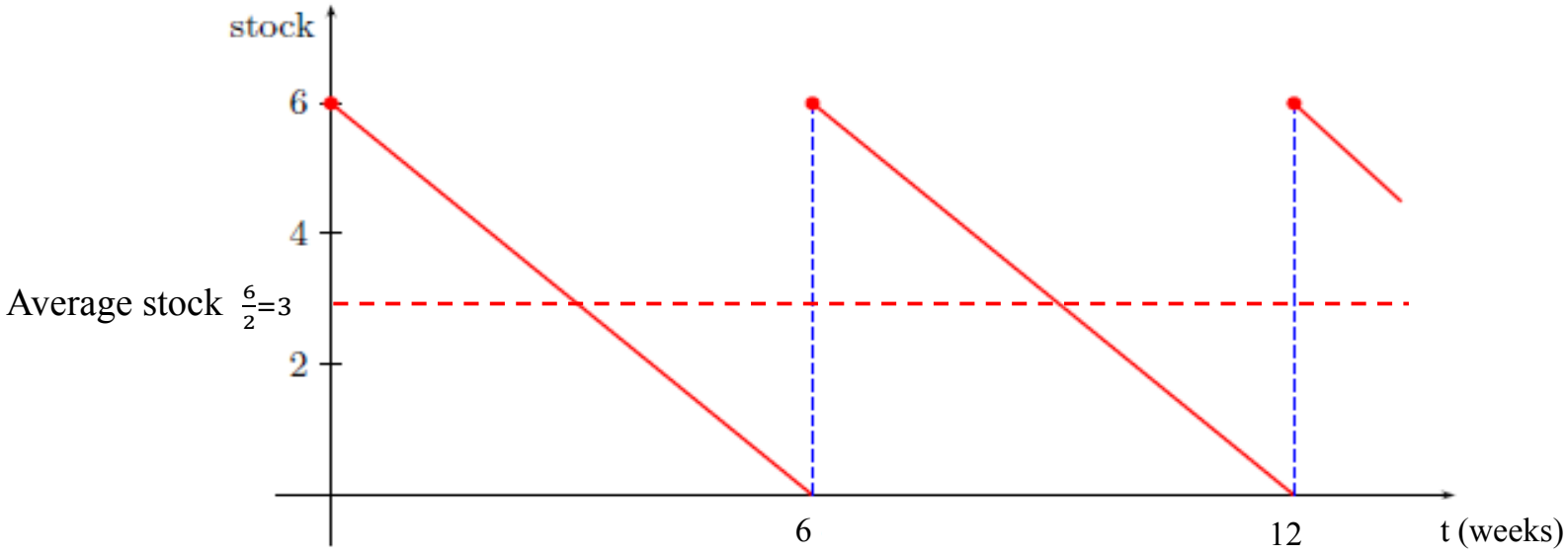
Inventory Example

- A car shop sells 1 car each week (say 50 per year)
- If shop runs out of cars, new ones are ordered and delivered immediately
- Every order costs 250 euros
- Stock costs are 4% of average value of cars in stock
- Each car has a value of 25000 euros

Question: How many cars should be ordered every time to minimize the total (ordering and stock) yearly costs?

Inventory Example - cont'd

- Suppose every time shop runs out of cars, 6 new ones are ordered
- Then, the development of stock over time is given as follows



Inventory Example - cont'd

- Let q denote the order quantity
- Given the 50 cars sold per year, the number of orders then is $\frac{50}{q}$
- Since each order costs 250 euros, the yearly ordering costs are

$$C_O(q) = 250 \cdot \frac{50}{q} = \frac{12500}{q}$$

- Note that the average stock level is $\frac{q}{2}$
- Since stock costs are 4% of value of average stock level, yearly stock costs are

$$C_S(q) = 25000 \cdot 0.04 \cdot \frac{q}{2} = 500q$$

- Yearly total costs are

$$C(q) = C_O(q) + C_S(q) = \frac{12500}{q} + 500q = \frac{12500 + 500q^2}{q}$$

Power

A function

$$f(x) = x^r$$

where r is some real number, is called a power function.

Some arithmetic rules (as a reminder)

- $x^a \cdot x^b = x^{a+b}$
- $x^{-a} = \frac{1}{x^a}$
- $x^0 = 1$
- $(x^a)^b = x^{ab}$
- $x^{\frac{1}{m}} = \sqrt[m]{x}$
- $1^r = 1$

Power - Example

- Power functions can model non-linear relationships between input and output.

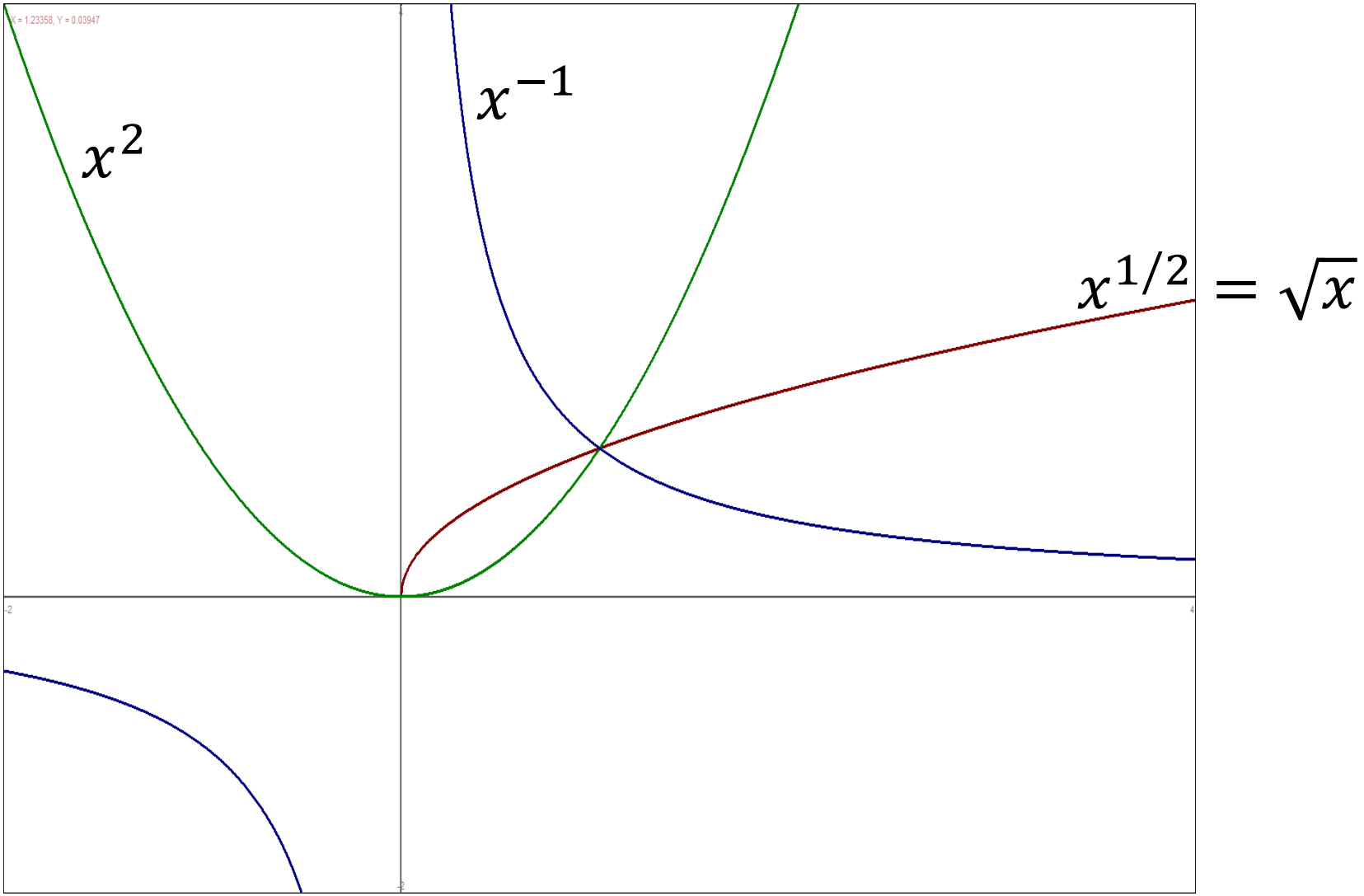
- **Example: Cobb-Douglas Production Function.** Total production (P) depends on labor (L) and capital (K) as follows:

$$P(L, K) = L^{\alpha} \cdot K^{\beta}$$

  If we hold capital (K) constant, the relationship between labor (L) and production (P) is a power function: $P(L) = L^{\alpha}$.

- The exponent α represents the *output elasticity* of labor, telling us how responsive production is to a change in labor.

Picturing Powers



Modeling Growth: Exponential Functions

- How model phenomena that grow at a constant percentage rate?

  $\rightarrow$ **exponential function** (heavily used in finance)

- **Example: Compound Interest**
  - Current balance: 1000 euros
  - Yearly interest rate: 5% (i.e., growth factor is 1.05)

If $f(t)$ denotes the balance after t years, the pattern is:

$$\begin{aligned} f(0) &= 1000 && = 1000(1.05)^0 \\ f(1) &= 1000 \cdot (1.05) && = 1000(1.05)^1 \\ f(2) &= (1000 \cdot 1.05) \cdot 1.05 && = 1000(1.05)^2 \end{aligned}$$

The general model for compound growth is $f(t) = P(1+r)^t$

Exponential

An exponential function is a function of the form

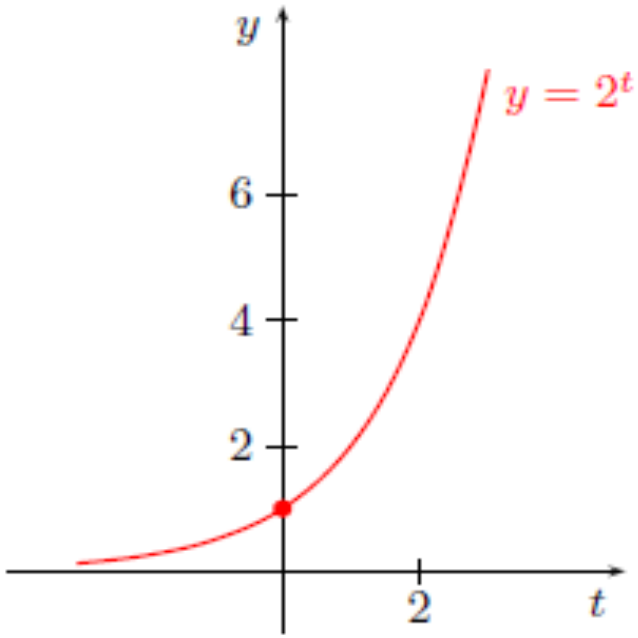
$$f(x) = a^x$$

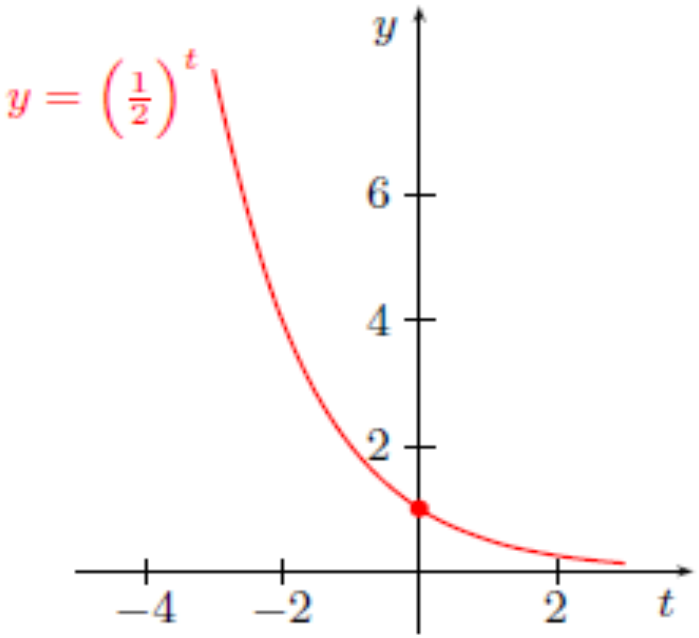
where $a > 0$ is the so-called base number.

Some arithmetic rules (as a reminder)

- $a^x \cdot a^y = a^{x+y}$
- $a^{-x} = \frac{1}{a^x}$
- $a^0 = 1$
- $(a^x)^y = a^{xy}$
- $(ab)^x = a^x b^x$
- $1^x = 1$

Picturing Exponentials





Bank Account Example - cont'd

Reconsider the bank account model: $f(t) = 1000(1.05)^t$

- A common question: How long does it take for our investment to double?

We need to solve for t in the equation:

$$2000 = 1000(1.05)^t \quad \Leftrightarrow \quad 2 = (1.05)^t$$

The problem: The variable we want to find, t, is in the exponent.

The solution: A tool that gets t out of the exponent, i.e., the **logarithm**.

$$t =^{1.05} \log 2 = \frac{\log 2}{\log 1.05} \approx 14.2 \text{ years}$$

Logarithm (Interlude)

The solution of the exponential equation

$$a^x = c$$

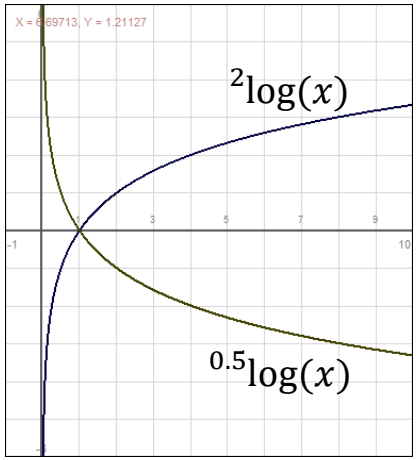
with $a > 0$ is given by

$$^a\log c = \frac{^{10}\log c}{^{10}\log a} = \frac{\log c}{\log a}$$

where we follow the usual convention of writing $\log b$ instead of $^{10}\log b$.

Arithmetic rules:

- $^a\log xy = {}^a\log x + {}^a\log y$
- $^a\log x^p = p \cdot {}^a\log x$
- $^a\log 1 = 0 \quad \text{and} \quad {}^a\log a = 1$



The Inverse Function

- We just used a logarithm to "undo" an exponential
- This concept of undoing a function is generalized by the **inverse function**
  - Not all operations can be undone
  - Not all functions have an inverse
  - But, if the inverse of a function f exists, we denote it as f^{-1}

Applications:

- A model $y = f(x)$ tells you the output y corresponding to input x. The inverse function ($x = f^{-1}(y)$) tells you the input needed to achieve a desired output.
- If $V(t)$ is the value of an investment after t years, then $V^{-1}(y)$ is the time required to reach a value of y.

Inverse - Examples

Some examples

$$f(x) = a^x \text{ has inverse } f^{-1}(y) =^a \log y$$

$$f(x) = x^2 \text{ has no inverse on } \mathbb{R}$$

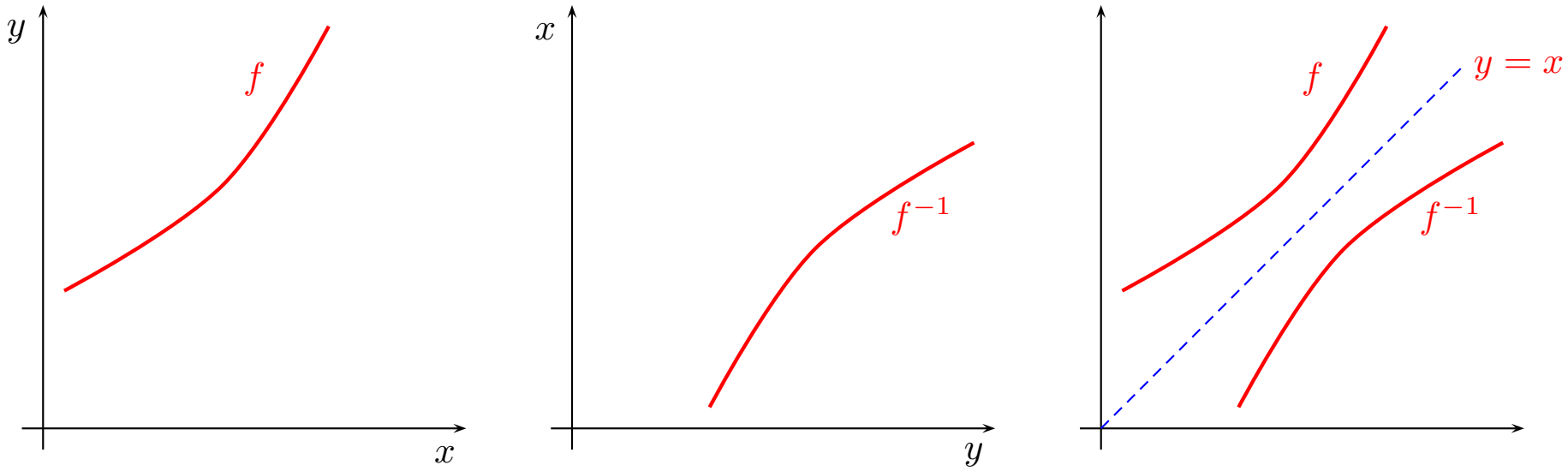
$$f(x) = x^2 \text{ confined to } \mathbb{R}_+ \text{ has inverse } f^{-1}(x) = \sqrt{x}$$

Picturing Inverses

THE GRAPHS OF A FUNCTION AND ITS INVERSE

The graphs of a function f and its inverse f^{-1} are symmetric about the line $y = x$.

This property is illustrated in Figure 8 below.



Computing Inverses

Inverting a linear function

$$y = 3x + 2 \quad \Leftrightarrow \quad y - 2 = 3x \quad \Leftrightarrow \quad \frac{1}{3}y - \frac{2}{3} = x \quad \Leftrightarrow \quad x = \frac{1}{3}y - \frac{2}{3}$$

Inverting a power function $(x, y \geq 0)$

$$y = x^r \quad \Leftrightarrow \quad (x^r)^{\frac{1}{r}} = (y)^{\frac{1}{r}} \quad \Leftrightarrow \quad x = y^{\frac{1}{r}}$$

Inverting a broken function $(x \neq 2,\ y \neq 1)$

$$y = \frac{x-1}{x-2} \quad \Leftrightarrow \quad y(x-2) = x - 1 \quad \Leftrightarrow \quad yx - 2y = x - 1 \quad \Leftrightarrow \quad yx - x = 2y - 1$$

$$\Leftrightarrow \quad x(y-1) = 2y - 1 \quad \Leftrightarrow \quad x = \frac{2y-1}{y-1}$$